The physical, chemical and mechanical properties of nanomaterials are based upon the structure of nanomaterials. This is because of the size of the particles being with in nanoscale (1-100 nm). *The two main factors for the difference in properties are due to increased relative surface area and quantum effects.*

With decrease in size of the particles, a greater proportion of the particles are found at the surface as compared two those in the bulk. As a result properties such as reactivity, strength and electrical etc. are changed or enhanced. A given mass of material in nanoparticle form will be much more reactive than the same mass of material made up of larger particles. It is because the growth processes and catalytic chemical reactions which occur at surfaces.

The change in properties from macro-scale to nano scale can be observed by taking a simple example of gold. If we start cutting gold from inches to centimeters from cm to mm and from mm to micros, we will notice the change in properties.

But when these μm size gold particles are further sliced into nano size particles, every thing will be changed including gold's colour, melting point and chemical properties. Melting point of nano gold is less than that of bulk gold melting point. Similarly instead of yellow colour, nano gold particles appear in different colour. This colour depends on the size of the particle. Not only for gold, all the materials will show the peculiar behaviour and change in their properties when they enter into the nano scale. That is why ***one nanometer is called as a magical point on the dimensional scale.***

One example is the "quantum size effect" where the electronic properties of solids are altered with great reductions in particle size. This effect does not come into play by going from macro to micro dimensions. However, it becomes pronounced when the nanometer size range is reached. A certain number of physical properties also alter with the change from macroscopic systems. Novel mechanical properties of nanomaterials are a subject of nanomechanics research. Catalytic activities also reveal new behaviour in the interaction with biomaterials. Thus manipulation of matter at nanoscale provides some novel properties to the material.

13.3 NANOCHEMISTRY AND APPLICATIONS

Nanochemistry is the utilization of synthetic chemistry to make nanoscale building blocks of different size and shape, composition and surface structure, charge and functionality. Its target is to control self assembly of these building blocks at various scale lengths.

Nanotechnology is concerned with a device, product or process, based upon individual or multiple integrate nanoscale components.

Applications of Nanochemistry

There are numerous applications of Nanochemistry in industry and research. Nanotechnology is a multi-disciplinary subject and it finds applications in almost all branches of science and engineering. Some important ones are listed below:

1. **Fabric:** The properties of familiar materials are being changed by manufacturers who are adding nano-sized components to conventional materials to improve performance. For example, some clothing manufacturers are making water and stain repellent clothing using nano-sized whiskers in the fabric that cause water to heat up on the surface.

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2. **Strength of materials:** Nano-sized particles of carbon, for example, nanotubes and bucky balls are extremely strong. Nanotubes and bucky balls are composed of only carbon and their strength comes from special characteristics of the bonds between carbon atoms. One proposed application that illustrates the strength of nanosized particles of carbon is the manufacture of T-shirt weight bullet proof vests made out of carbon nanotubes.

3. **Drug delivery:** Today, most harmful side effect of treatments such as chemotherapy is a result of drug delivery methods that don't pinpoint their intended target cell accurately. Researchers at Harvard and MIT have been able to attach special RNA strands, measuring about 10 nm in diameter, to nanoparticles and fill the nanoparticles with a chemotherapy drug. These RNA strands are attracted to cancer cells. When the nanoparticle encounters a cancer cell it adheres to it and releases the drug into the cancer cell. This directed method of drug delivery has great potential for treating cancer patients while producing less harmful side effects than those produced by conventional chemotherapy.

4. **Micro Nano Electromechanical Systems:** The ability to create gears, mirrors, sensor elements as well as electronic circuitry in silicon surfaces allows the manufacture of miniature sensors such as those used to activate the airbags in the cars. This technique is called MEMS (Micro-Electro Mechanical Systems). The MEMS technique results in close integration of the mechanical mechanism with the necessary electronic circuit on a single silicon chip, similar to the method used to produce computer chips. Using MEMS to produce a device reduces both the cost and size of the product, compared to similar devices made with conventional methods. MEMS is a stepping stone to NEMS or *Nano-Electro-Mechanical Systems*. NEMS products are being made by a few companies, and will take over as the standard once manufacturers make the investment in the equipment needed to produce nano-sized features.

5. **Molecular Manufacturing:** Researchers are working on developing a method called molecular manufacturing that may be called a *molecular fabricator*; this device would use tiny manipulators to position atoms and molecules to build an object as complex as a desktop computer. Researchers believe that raw materials can be used to reproduce almost any inanimate object using this method.

Two main approaches are used in nanotechnology. (*a*) Bottom-up approach (*b*) Top-down approach. In the "bottom-up" approach (synthesizing), materials and devices are built from molecular components which assemble themselves chemically by principles of molecular recognition.

Bottom-up approaches, uses the chemical properties of single molecules to cause single-molecule components to (*a*) self-organize or self-assemble into some useful conformation, or (*b*) rely on positional assembly. These approaches utilize the concepts of molecular self-assembly and/or molecular recognition. Such bottom-up approaches should, broadly speaking, be able to produce devices in parallel and much cheaper than top-down methods, but could potentially be overwhelmed as the size and complexity of the desired assembly increases.

In the "top-down" approach (fabricating), nano-objects are constructed from larger entities without atomic-level control. This approach is analogous to making a statute from the block of marble and is considered as natural extension of current methods of microelectronics in which structures of limited dimensions are created by laying down thin layers of material and etching away unwanted parts of each layer. It uses the traditional workshop or micro-fabrication method where externally controlled tools are used to cut, mill, and shape materials into the desired shape and order.

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6. **Miscellaneous Applications:**
   (*a*) It is used in formation of
   (*i*) Nanooxides of nobel, ferromagnetic and rare metals like Ag, Au, Ti, Zr etc.
   (*ii*) Nanomembranes and nanopolymers.
   (*iii*) Nanomaterials such as cement, fertilizer, nanopowders, toothpaste, etc.
   (*iv*) Cosmetics, e.g. sunscreen, skincare products, etc.
   (*v*) Screens and photographic films.
   (*vi*) Exhaust purification.
   (*vii*) Built in ovens and baking trays.
   (*viii*) Catalyst for chemical reactions.
   (*b*) If finds applications in the following fields:
   (*i*) Nano energy application.
   (*ii*) Nanogreen chemistry.
   (*iii*) Separating technology for waste water treatment and food production.

13.4 BOTTOM-UP AND TOP-DOWN APPROACHES IN NANO TECHNOLOGY

Nanostructures or particles can be created by either of the two approaches as shown in Fig. 13.1. A tabulated comparison of two approaches gives a summary of the two strategies.

13.4.1 Comparison of the Top Down and Bottom Up Processes

	Top Down Process	*Bottom Up Process*
(*i*)	It involves slicing or successive cutting of a bulk material to get nano sized particles.	Process involves building up of a material from the bottom; atom by atom, molecule by molecule or cluster by cluster.
(*ii*)	Attrition or Milling is a typical top down method in making nano particles.	Colloidal dispersion is an example of bottom up approach in the synthesis of nano particles.
(*iii*)	Problem of imperfection of surface structure, internal stress and significant crystallographic damage to the processed patterns leading to design and fabrication problems.	Absence of available tools at nano scale favours bottom up approach which provides a better chance to obtain nano structures with less defects, more homogeneous chemical composition.

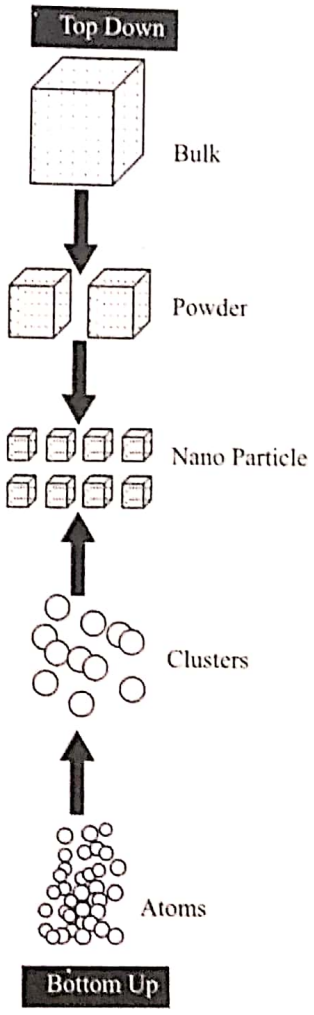


Fig. 13.1 Schematic representation of the building up of Nanostructures.

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